02-Commands-and-Configuration.md

SIEMLab — Commands & Configuration Reference

> This document contains every command and configuration file used during the lab build, organized by host and by phase. Use it alongside `01-SIEMLab-Architecture-and-Setup.md` for the reasoning behind each step.

A. VMware — Network Setup

Creating the custom Host-only network:

1. Edit → Virtual Network Editor → Add Network → select `VMnet2`
2. Type: **Host-only**
3. Uncheck "Connect a host virtual adapter"
4. Uncheck "Use local DHCP service"
5. Subnet: `192.168.56.0/24`

Static IP allocation table:

Host	Interface	IP	Gateway	DNS
DC	Host-only	192.168.56.10	—	192.168.56.10
DC	NAT	DHCP-assigned	DHCP-assigned	192.168.56.10 (manual)
Win10	Host-only	192.168.56.20	—	192.168.56.10
Win10	NAT	DHCP-assigned	DHCP-assigned	192.168.56.10 (manual)
Ubuntu	Host-only	192.168.56.30	—	192.168.56.10
Ubuntu	NAT	DHCP-assigned	DHCP-assigned	192.168.56.10 (no override)

B. Windows Server (DC) — PowerShell Commands

B.1. Rename the host & set the static IP (via GUI or PowerShell)

```
Rename-Computer -NewName "DC01" -Restart
```

B.2. Install the AD DS role and promote to Domain Controller

```
Install-WindowsFeature AD-Domain-Services -IncludeManagementTools

Install-ADDSForest `
  -DomainName "cslab.local" `
  -DomainNetbiosName "CSLAB" `
  -InstallDns:$true `
  -Force:$true
```

B.3. Configure the DNS forwarder (external resolution)

```
Add-DnsServerForwarder -IPAddress "8.8.8.8", "1.1.1.1"

# Verify
Get-DnsServerForwarder
```

Or via GUI: **Server Manager** → **Tools** → **DNS** → **right-click server** → **Properties** → **Forwarders tab** → **Edit** → **add 8.8.8.8, 1.1.1.1**

B.4. Disable Dynamic DNS Registration on the NAT interface

GUI: **Network Adapter Properties (NAT interface)** → **IPv4 Properties** → **Advanced** → **DNS tab** → **uncheck "Register this connection's addresses in DNS"**

B.5. Full network configuration check

```
ipconfig /all
```

B.6. Verify the routing table (confirm a single default route via NAT)

```
route print
```

B.7. Test DNS resolution + internet connectivity

```
# Resolve an external domain (via forwarder)
nslookup google.com

# Resolve an internal domain
nslookup cslab.local

# Test actual internet reachability (not just DNS)
Test-NetConnection google.com -Port 443
```

```
# Confirm reachability to self via the internal interface
Test-NetConnection 192.168.56.10 -Port 445
```

C. Windows 10 Client — PowerShell Commands

C.1. Set the static IP + DNS (via GUI or PowerShell)

```
New-NetIPAddress -InterfaceAlias "Ethernet0" -IPAddress "192.168.56.20" -
PrefixLength 24
Set-DnsClientServerAddress -InterfaceAlias "Ethernet0" -ServerAddresses
"192.168.56.10"
```

C.2. Join the domain

```
Add-Computer -DomainName "cslab.local" -Restart
```

C.3. Install Sysmon with the SwiftOnSecurity configuration

```
mkdir C:\Tools
cd C:\Tools

# Download Sysmon from Microsoft Sysinternals
Invoke-WebRequest -Uri "https://download.sysinternals.com/files/Sysmon.zip"
-OutFile "Sysmon.zip"
Expand-Archive -Path "Sysmon.zip" -DestinationPath "Sysmon"

# Download the SwiftOnSecurity configuration
Invoke-WebRequest -Uri
"https://raw.githubusercontent.com/SwiftOnSecurity/sysmon-
config/master/sysmonconfig-export.xml" -OutFile "sysmonconfig.xml"

# Install Sysmon with the configuration
cd C:\Tools\Sysmon
.\Sysmon64.exe -i C:\Tools\sysmonconfig.xml -accepteula

# Verify the service
Get-Service sysmon64
```

View logs at: `Event Viewer → Applications and Services Logs → Microsoft → Windows → Sysmon → Operational`

C.4. Install the Splunk Universal Forwarder

Download the `.msi` installer from: `https://www.splunk.com/en_us/download/universal-forwarder.html`

During installation (GUI installer):

- **Receiving Indexer:** `192.168.56.30:9997`
- **Deployment Server:** `192.168.56.30:8089`

C.5. Repointing the Deployment Server after install (without re-running the installer)

```
cd "C:\Program Files\SplunkUniversalForwarder\bin"
.\splunk.exe set deploy-poll 192.168.56.30:8089
.\splunk.exe restart
```

C.6. Verify the forwarder received the app from the Deployment Server

```
dir "C:\Program Files\SplunkUniversalForwarder\etc\apps\sysmon_inputs\local"
```

C.7. Check the forwarder log when troubleshooting connection issues

```
Get-Content "C:\Program Files\SplunkUniversalForwarder\var\log\splunk\splunkd.log" -Tail 50
```

D. Ubuntu Server — Bash Commands

D.1. Check interfaces & set the static IP via Netplan (Host-only)

```
ip a
sudo nano /etc/netplan/00-installer-config.yaml
```

```
network:
  version: 2
  ethernets:
    ens33:                          # host-only interface name — adjust to match your system
      dhcp4: no
      addresses:
        - 192.168.56.30/24
```

```
      nameservers:
        addresses: [192.168.56.10]
```

```
sudo netplan apply
ip a
ping 192.168.56.10
```

D.2. Add the NAT interface (dual-homing) — update Netplan

```
network:
  version: 2
  ethernets:
    ens33:                          # host-only
      dhcp4: no
      addresses:
        - 192.168.56.30/24
      nameservers:
        addresses: [192.168.56.10]
    ens37:                          # NAT — interface name may vary by
system
      dhcp4: yes
      dhcp4-overrides:
        use-dns: false              # keep DNS pointed at the DC; don't let
NAT override it
```

```
sudo netplan apply
```

D.3. Verify routing (only one default route, via NAT)

```
ip route
```

Expected output: